

Xinran (Nicole) Han

✉ xinranhan@g.harvard.edu | 🏠 xrhan.github.io

Education

Harvard University

PH.D. CANDIDATE IN COMPUTER SCIENCE

Cambridge, Massachusetts, USA

Sept. 2020 - 2026 (Expected)

- Advisor: Prof. Todd Zickler. Master of Science in Computer Science *en route* (2022), GPA: 4.00/4.00.

University of Pennsylvania

BACHELOR OF SCIENCE IN ENGINEERING

Philadelphia, Pennsylvania, USA

Sept. 2016 - May 2020

- Double Major in Computer Science and Mathematics, *Summa Cum Laude*. GPA: 4.00/4.00

Research Interests

My research lies at the intersection of generative models, computer vision, and multimodal learning. I develop efficient architectures, representations and inference-time algorithms for visual perception and reasoning, drawing on insights from mathematical modeling and human visual perception. More broadly, I aim to build perceptually grounded visual systems that generalize robustly and learn efficiently.

Selected Publications

Generative Perception of Shape and Material from Differential Motion

2025

XINRAN HAN, KO NISHINO, TODD ZICKLER

- Advances in Neural Information Processing Systems (NeurIPS) 2025.
- Designed and trained a conditional video diffusion model from scratch to jointly disentangle 3D shape and materials from short clips using motion, demonstrating generative modeling as a promising approach for ambiguity-aware world modeling.

Multistable Shape from Shading Emerges from Patch Diffusion

2024

XINRAN HAN, TODD ZICKLER, KO NISHINO

- Advances in Neural Information Processing Systems (NeurIPS) 2024, **Spotlight Paper** (Top 2 %).
- Developed a small patch-based diffusion model (10MB in size) with test-time scaling for ambiguity-aware geometry inference, generating diverse structured predictions aligned with human perception and surpassing larger foundation models.

Transfer Learning for Bayesian Optimization on Heterogeneous Search Spaces

2024

ZHOU FAN, XINRAN HAN AND ZI WANG.

- Transactions on Machine Learning Research (TMLR) 2024. Research collaboration with Google Brain.

Curvature Fields from Shading Fields

2023

XINRAN HAN AND TODD ZICKLER.

- Proceedings of the NeurIPS Workshop on Symmetry and Geometry in Neural Representations, PMLR 228, 2023.
- Derived an equivariant neural field for inferring lighting- and texture-invariant curvature representation from images.

Compositional Data and Task Augmentation for Instruction Following

2021

SOHAM DAN*, XINRAN HAN* AND DAN ROTH. (* EQUAL CONTRIBUTIONS)

- Findings, Conference on Empirical Methods in Natural Language Processing (EMNLP) 2021.
- Proposed compositional data augmentation and auxiliary spatial-reasoning tasks for grounded instruction following, improving a multimodal agent's performance by up to 14% in low-data settings.

ForkGAN: Seeing into the Rainy Night

2020

ZIJIANG ZHENG, YANG WU, XINRAN HAN AND JIANBO SHI

- European Conference on Computer Vision (ECCV) 2020, **Oral Presentation** (Top 2 %).
- Introduced a disentangled image translation model for domain adaptation and robust visual perception under adverse conditions.

Experience

Apple Inc.

Cupertino, CA

RESEARCH INTERN

Summer 2025

- **Consistent Video Generation with Representation Alignment** [Project Page]
- Extended and finetuned a **5B-parameter image/text-to-video diffusion model** for timelapse video generation, scaling training to multi-node H100 clusters with DeepSpeed.
- Built a scalable **VLM-based data curation pipeline** to automatically filter high-quality training videos from mixed data sources.
- Introduced **derivative representation alignment (dREPA)**, a method that aligns video generation with video understanding models in latent space, improving subject consistency over long-horizon generation and generalization across artistic styles.

Computer Vision Lab, Kyoto University

Kyoto, Japan

VISITING RESEARCHER

Summer 2023, 2024

- Collaborated with Prof. Ko Nishino on generative models for 3D vision and visual reasoning. Proposed a generative perception framework bridging diffusion models with human visual perception, resulting in first-author papers at NeurIPS 2024 and NeurIPS 2025.

Cognitive Computation Group, University of Pennsylvania

Philadelphia, PA

RESEARCH ASSISTANT

2019 - 2020

- Worked on grounded language understanding for compositional instruction-following agents, with publication at EMNLP 2021.
- Built open-source tools for Wikipedia processing and multilingual supervision data pipeline. [Code]

GRASP Lab, University of Pennsylvania

Philadelphia, PA

RESEARCH ASSISTANT

2017 - 2020

- Worked on GAN-based cross-domain image translation for autonomous driving (ECCV 2020 Oral) and multimodal representation learning across vision and audio.

AQR Capital Management

Greenwich, CT

QUANTITATIVE RESEARCHER INTERN

Summer 2019

- Global Alternative Premia Team

Technical Skills

- **Generative & Multimodal Model Training:** Trained diffusion-based image and video models from scratch with custom architecture designs, and post-trained 5B-parameter multimodal video models. Experienced with mixed-precision training, DDP, DeepSpeed, and VLM-based data curation.
- **Languages & Tools:** Python, C++, PyTorch, Mitsuba3

Awards & Honors

2025-2026 **Siebel Scholar**, Class of 2026 (awarded to top graduate students in Computer Science)

2020-2021 **Mou-Shiung Lin Fellowship**, Harvard University

2020 **Hugo Otto Wolf Memorial Prize**, University of Pennsylvania (two senior students per department)

2016-2020 **Dean's List**, University of Pennsylvania

2015 **Team China Selection Team**, International Young Physicists' Tournament (IYPT)

2014 **First Prize**, Chinese Physics Olympiad (CPhO)

Invited Talks

2026.2	Perception as Generation: Navigating Ambiguity with Diffusion Models , Harvard University, Computer Science Colloquium	Boston, MA
2025.5	Computational Models Exhibit Invariance and Multistability In Shape from Shading , Vision Sciences Society (VSS) Annual Meeting	Tampa, FL
2025.1	Towards Aligning Human and Computer Shape Perception , Boston University, Artificial Intelligence Research (AIR) Seminar	Boston, MA
2024.6	Multistable Shape from Shading Emerges from Patch Diffusion , Kyoto University, Cognitive Informatics Lab Seminar	Kyoto, Japan
2023.12	Curvature Fields from Shading Fields , New England Computer Vision Workshop (NECV)	Hanover, NH

Teaching & Services

Teaching

TEACHING ASSISTANT

- **ES 143 Computer Vision**, Spring 2024. **CS 283 Advanced Computer Vision**, Fall 2021, Harvard University.
- **CIS 160 Mathematical Foundations of Computer Science**, Fall 2017 & Spring 2018, University of Pennsylvania.

Academic service

REVIEWER

- AISTATS 2023, ECCV 2024, NeurIPS 2024 & 2025, CVPR 2025 & 2026, ICCV 2025